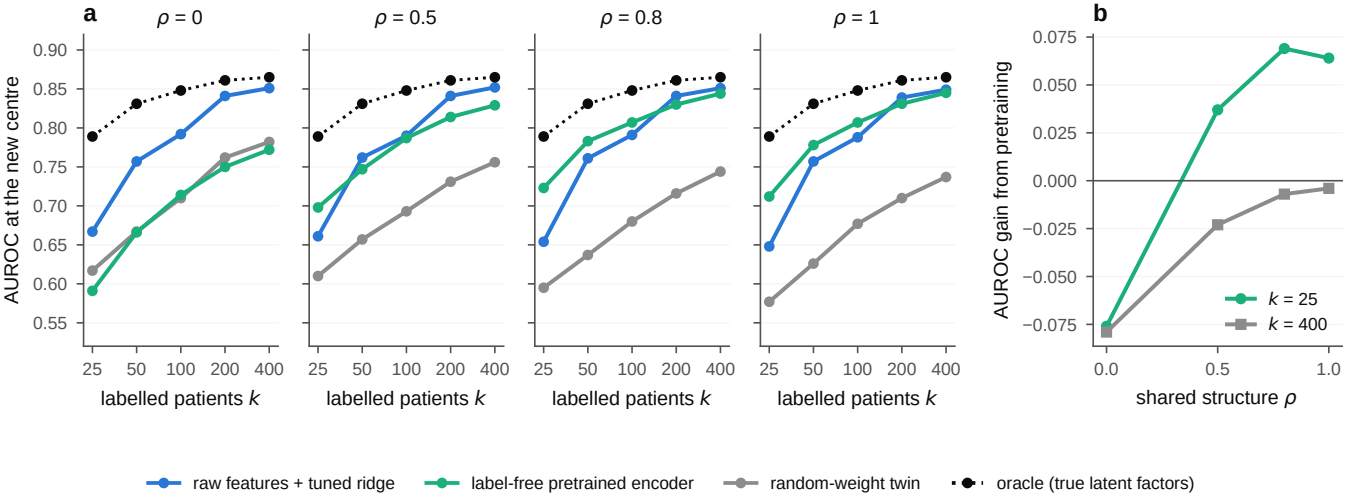
a
ρ = 0
ρ = 0.5
ρ = 0.8
ρ = 1
AUROC at the new centre
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
25 50 100 200 400
labelled patients k
b
AUROC gain from pretraining
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
k = 25
k = 400
0.0 0.5 1.0
shared structure ρ
raw features + tuned ridge
label-free pretrained encoder
random-weight twin
oracle (true latent factors)